

Front wing in ground effect beside a rotating wheel

Steady RANS setup and verification report

OpenFOAM v2606 (native) • `snappyHexMesh` + `simpleFoam` • case S1.42_h0.13_AOA8_W0.63

1 Case at a glance

Configuration		Mesh	
wing half-span	$S/c = n/a$	cells	n/a
ride height	$h/c = n/a$	background cells	n/a
angle of attack	n/a°	layer coverage	96.9 %
wheel width	$W/c = n/a$	<code>checkMesh</code> clean	no
Flow		Cost, 20 MPI ranks	
freestream	n/a m/s	meshing	n/a min
chord	n/a m	solving	n/a min
Re_c	n/a	per iteration	n/a s
ρ	n/a kg/m ³	iterations	n/a
ν	n/a m ² /s	converged	no

Headline result. Force *coefficient-areas*, mean over the last 50 iterations. $A_{\text{ref}} = 1 \text{ m}^2$ on purpose, so these are $C_L \cdot A$ and $C_D \cdot A$ in m² — the quantity the Diasinos papers tabulate. Negative lift is downforce.

	$C_L \cdot A$ [m ²]	$C_D \cdot A$ [m ²]	spread of $C_L \cdot A$
wing	n/a	n/a	n/a %
wheel	n/a	n/a	n/a %
total	n/a	n/a	n/a %

Mean y^+ : wing n/a, wheel n/a, maximum n/a.

2 Software and reproducibility

- **OpenFOAM v2606** (ESI/OpenCFD), built from source. Native only — no commercial solver, no in-house libraries.
- Mesh `snappyHexMesh`, solve `simpleFoam` (segregated, SIMPLEC), turbulence `kOmegaSST`.
- Geometry generated parametrically from the Diasinos specification; 420 configurations available.
- Everything else is `python3` standard library. No `numpy/pandas` needed to run a case.

To reproduce this case.

```
cp env/project.env.example env/project.env # set OPENFOAM_BASHRC and NP
cd Diasinos_Geometry_Generator && .venv/bin/python -m geometry_generator.generate
cd ../RANS_Simulations
python3 prepare_case.py --span 1.42 --height 0.13 --aoa 8 --width 0.63
cd S1.42_h0.13_AOA8_W0.63 && ./Allrun
```

3 Geometry

- Coordinate system: x streamwise (wheel centre at $x = 0$), y spanwise outboard, z up with the ground at $z = 0$. Half model, symmetry at $y = 0$.
- Chord $c = 75$ mm. Inverted NACA 4412, blunt trailing edge 0.5 mm. Wheel diameter $1.17c$, shoulder radius $0.067c$.
- **The generator writes millimetres**; everything in the simulation is metres. **Allmesh** scales by 0.001 on the way in.
- The *plinth* is a collar that deliberately pierces the ground plane (down to $z = -6$ mm) so the contact patch is a clean cut instead of a tangential singularity.

wing + endplate
wheel
plinth (cuts the ground)
ground $z = 0$, belt at 10 m/s

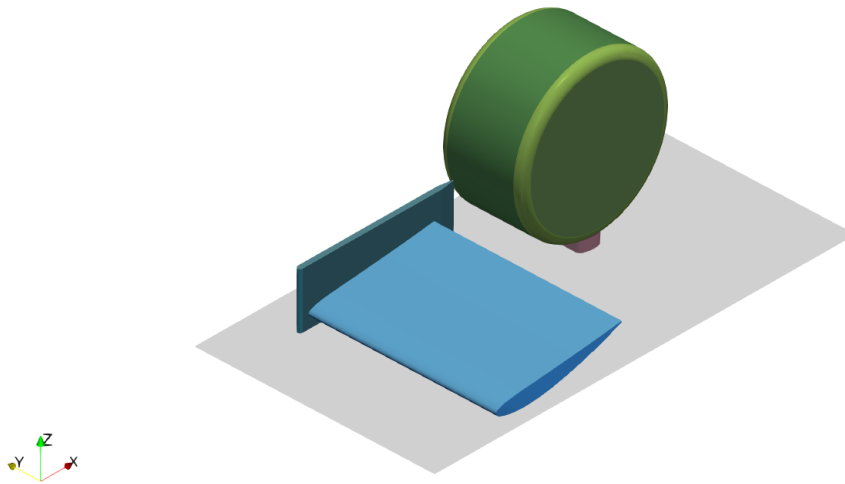


Figure 1: Geometry and the ground plane. The wing sits in ground effect ahead of and inboard of the wheel; the plinth passes through the ground.

Known geometry limitations.

- The wing is **prismatic**, extruded between two spanwise stations. Geometrically exact for a constant section; it does mean the triangles have a very high aspect ratio, which **surfaceCheck** reports as poor quality. Not an error.
- **Two duplicate triangles** on the endplate outer edge — a defect in the generator. Two facets out of 1142; **snappyHexMesh** tolerates them. Reported by **Allmesh** and deliberately non-fatal.
- Both STLs consist of two disjoint parts (wing + endplate, wheel + plinth), so **surfaceCheck** says “not closed”. Expected: no **mode inside** refinement region is attached to either body.
- Wheel tessellation was raised to 900 circumferential segments (0.31 mm facets) to match the 0.293 mm cell. At the original 180 segments the facets were 1.53 mm and the layers on the tread were squeezed by them.

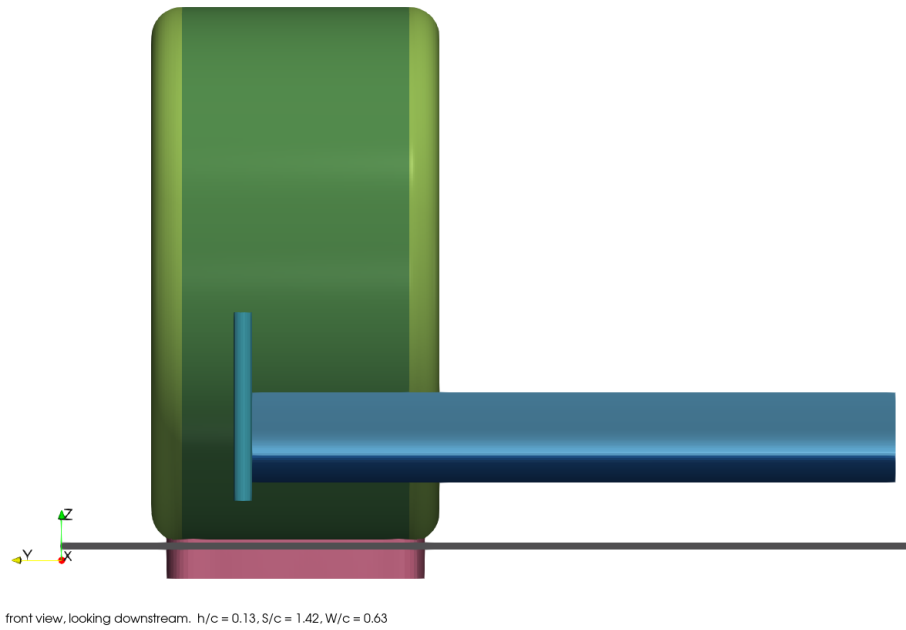


Figure 2: Front view, looking downstream. The endplate (light blue) falls inside the wheel silhouette: at $S/c = n/a$ and $W/c = n/a$ the spanwise gap is $-0.45c$, i.e. wing and wheel overlap.

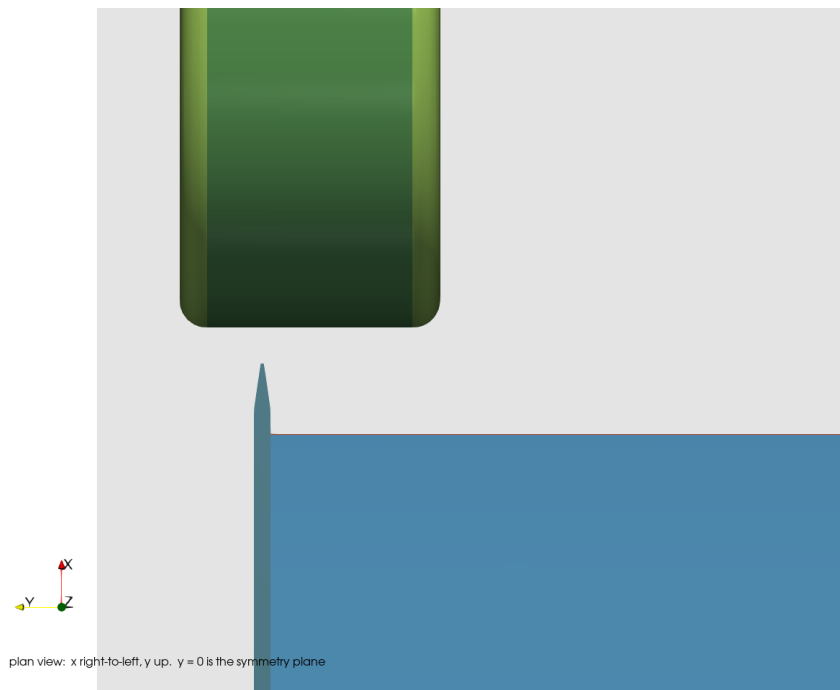


Figure 3: Plan view. $y = 0$ is the symmetry plane.

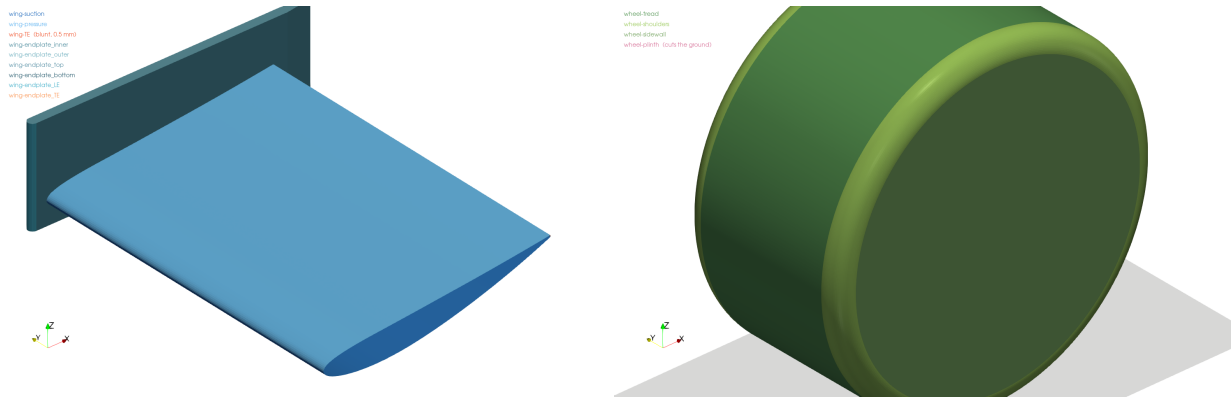


Figure 4: Named STL regions. `snappyHexMeshDict` refers to these names directly to set refinement level and patch name, so renaming a region in the generator breaks the mesh setup.

domain 1.894 x 0.675 x 0.638 m = 25.3c x 9c x 8.5c
 grey = ground (moving wall) yellow = symmetry plane y=0

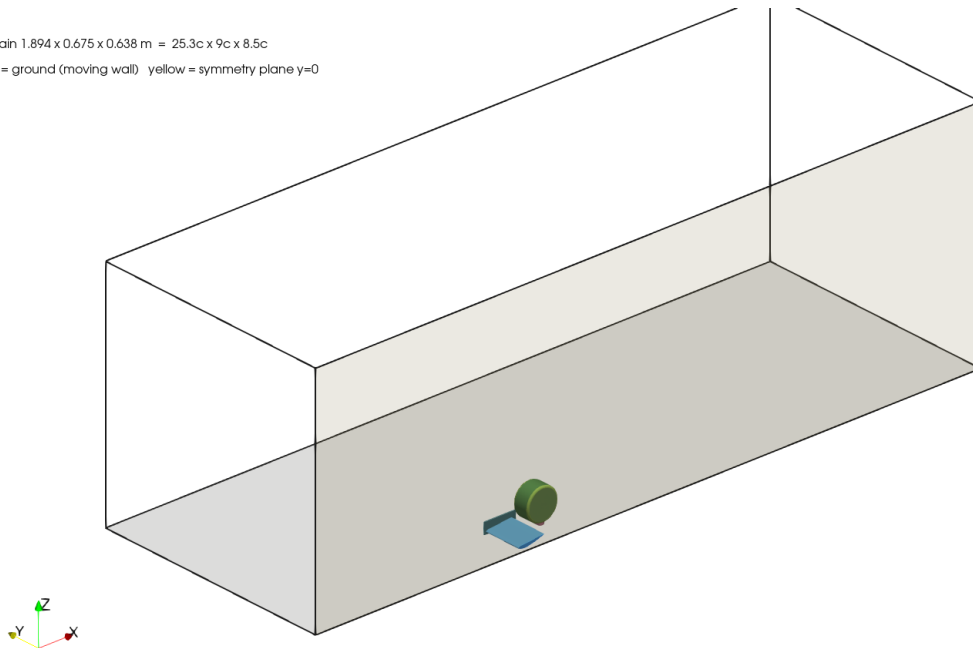


Figure 5: Computational domain, $1.894 \text{ m} \times 0.675 \text{ m} \times 0.638 \text{ m} = 25.3c \times 9c \times 8.5c$. Grey: moving ground. Yellow: symmetry plane.

4 Domain and boundary conditions

patch / group	type	U	p	k, ω, ν_t
inlet	patch	fixedValue (1000)	zeroGradient	turbulent inlet
outlet	patch	pressureInletOutletVelocity	fixedValue 0	inletOutlet
symmetry	symmetry	symmetry	symmetry	symmetry
side, sky	patch	slip	slip	slip / calculated
ground	wall	fixedValue (1000)	zeroGradient	wall functions
wingGroup	wall	noSlip	zeroGradient	wall functions
wheelRotating	wall	rotatingWallVelocity	zeroGradient	wall functions
wheelGround	wall	fixedValue (1000)	zeroGradient	wall functions

- **Rolling road:** the belt translates downstream at the freestream speed, so the ground is at rest relative to the far-field air.
- **Rotating wheel:** $\omega = -227.92$ rad/s about (0 1 0) through (0, 0, 0.043875 m). `rotatingWallVelocity` imposes $\omega \times r$ without moving the mesh — correct for steady RANS. The sign follows from requiring the contact point to move with the belt: $\omega_y = -U_\infty/R$. Reversing it spins the wheel backwards and still converges happily.
- The **wheel-plinth** patch is in **wheelGround**, not **wheelRotating**: it is the collar through the ground and moves with the belt.
- **side** and **sky** are **patch**, not **wall**, on purpose — a wall there would enter the **meshWave** wall-distance field that **kOmegaSST** blends on, and would appear in the force groups.

5 Mesh

- Background: $101 \times 36 \times 34$ **cubic** cells of 18.75 mm ($= 0.25c$), n/a cells. Cubic matters: each refinement level halves the edge, so the level-to-size table only holds from a cube.
- Level sizes: L4 1.17 mm, L5 0.586 mm, L6 0.293 mm, L7 0.146 mm.
- Per-region surface levels, distance refinement around both bodies (L5 within 15 mm, L4 within 30 mm), and nine **searchableBox** zones.
- Final mesh n/a cells.

refinement level	cells	share [%]
L0	110 876	0.6
L1	19 040	0.1
L2	501 184	2.7
L3	572 976	3.1
L4	4 188 141	22.8
L5	11 148 236	60.8
L6	820 502	4.5
L7	971 698	5.3

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Figure 6: Mesh on a spanwise slice through the wheel centre. The nested refinement zones and the wake box are visible.

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Figure 7: Same slice, wing trailing edge to wheel front — the region the whole study is about.

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Figure 8: Slice through the wheel axis ($x = 0$). Looking downstream, so y increases to the left.

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Figure 9: Mesh 1.2 mm above the belt. The contact-patch refinement box is the fine square under the wheel.

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Figure 10: Wheel tread at the top. Prism layers on the 0.31 mm-tessellated tread.

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Figure 11: Wheel tread, upstream face.

figure mesh_layers_shoulder.png not generated

Figure 12: Wheel shoulder, inboard. The 5 mm shoulder radius is the tightest convex curvature on the wheel.

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Figure 13: Contact patch. The plinth cuts the ground plane at $z = 0$; this is where the layers fight each other.

figure mesh_wing_te.png not generated

Figure 14: Blunt trailing edge, 0.5 mm thick. No prism layers — see below.

5.1 Prism layers

Four layers from 0.14 mm with expansion 1.3 (total 0.87 mm) on wing and wheel, three from 0.25 mm on the ground. This targets $y^+ \approx 2.5$, which is why the wall functions are the all- y^+ variants.

patch	faces	layers wanted	layers built	thickness [%]
not available				

Reading the layer table.

- Overall coverage 96.9 %.
- The patches in **bold** got **no layers at all**. These are **wing-TE**, **wing-endplate_TE** and **wheel-plinth** — sharp convex edges and the plinth/ground intersection. Asking for fewer and thinner layers does not change it: the limit is topology, not thickness.
- **This is accepted.** **snappyHexMesh** is known for incomplete layer coverage at sharp edges and where two layered surfaces meet. All three patches sit in L7 cells (0.146 mm), so the near-wall spacing is comparable to the layered regions anyway.
- The number to judge the near-wall treatment by is y^+ (section 8), not the layer count.

5.2 Mesh quality

Quality limits are deliberately relaxed (**maxNonOrtho** 75, **maxInternalSkewness** 6, **minTetQuality** -10^{30}), carried over from the previous HELYX setup. Enforcing a positive tet quality with absolute layer thicknesses on curved surfaces costs most of the layer coverage.

checkMesh clean: **no** (0 findings).

- no failures reported

These are the expected consequence of the relaxed limits. The solver runs stably on this mesh; if it ever diverges, **meshQualityDict** is the first thing to question.

6 Physics and numerics

turbulence model	kOmegaSST
ν_t wall function	nutUSpaldingWallFunction (continuous all y^+)
k wall function	kLowReWallFunction
ω wall function	omegaWallFunction
momentum convection	bounded Gauss linearUpwindV grad(U)
turbulence convection	bounded Gauss upwind (first order, deliberate)
gradients	cellLimited Gauss linear 1
laplacian	Gauss linear corrected
wall distance	meshWave
pressure-velocity	SIMPLEC (consistent yes)
p solver	GAMG, tolerance 10^{-7} , relTol 0.01
U, k, ω	smoothSolver GaussSeidel
relaxation	U 0.9, k and ω 0.7
residualControl	p, U : 10^{-5} ; k, ω : 10^{-4}
initialisation	potentialFoam, divergence-free start

Turbulence convection is first order on purpose. Second-order k/ω transport is the usual reason residuals stall on a mesh of this kind, and the effect on the integrated forces is small. Momentum convection is second order.

Differences against the previous HELYX setup. Relevant whenever the two are compared:

- **No curvature correction** (HELYX used Menter–Smirnov). The endplate and wheel vortex cores will diffuse somewhat faster.
- **No separation fix** (HELYX used the Rumsey form). Separation onset on the suction side may differ.
- The adaptive wall treatment *is* reproduced, by the all- y^+ wall functions above.
- HELYX ran a coupled solver pseudo-transiently; this is segregated and steady, so iteration counts are not comparable.

7 Convergence

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Figure 15: Initial residuals per SIMPLE iteration, with the `residualControl` thresholds marked.

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Figure 16: Force history. The grey band is the window averaged for the reported result.

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Figure 17: Settling of the total $C_L \cdot A$ over the final iterations.

How to judge this.

- `residualControl` satisfied: **no**.
- Final initial residuals: p n/a, U_x n/a, U_z n/a, k n/a, ω n/a.
- **The force spread matters more than the residual.** A converged residual with $C_L \cdot A$ still swinging by several percent means an unsteady flow being forced into a steady solve. Here the total spread over the averaging window is n/a %.
- Residual *fields* are written as well, so a stalled case can be opened in ParaView to see *where* it will not settle.

8 Results

9 Assessment

What supports trusting these numbers.

- **Background mesh** exactly as designed (n/a cells), cubic, dividing the domain without remainder.
- **Refinement** reproduces the previous validated HELYX setup level for level, and the final cell count agrees with it closely.

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Figure 18: Pressure coefficient on the wing suction side.

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Figure 19: Total pressure coefficient on streamwise slices — vortex cores and wake losses.

- Patch groups resolve correctly — `patchSummary` confirms `noSlip` on the wing, `rotatingWallVelocity` on the wheel and the belt velocity on the plinth.
- Wheel rotation sign verified analytically against the belt speed.
- Divergence-free initialisation via `potentialFoam`.
- y^+ in the design range of the chosen wall functions.

What to be careful about.

- No curvature correction and no separation fix relative to HELYX. Expect faster vortex diffusion and possibly different separation onset.
- Relaxed mesh quality limits leave genuinely poor cells in the mesh (`minTetQuality` disabled). Accepted in exchange for layer coverage.
- Three patches without prism layers, see section 5.
- Two duplicate triangles in the wing STL.
- Steady RANS on a flow that may not be steady. At high angle of attack or low ride height, check the force spread before quoting a number.
- The two local refinement boxes that stand in for HELYX's per-region distance refinement are positioned for this geometry and do not follow the parameters. Re-check them before extending the sweep.

Generated from S1.42_h0.13_AOA8_W0.63. Figures by `pvbatch` (ParaView) and `matplotlib`; numbers by `report/scripts/make_tex_` from the case's `results.json` and solver logs.

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Figure 20: y^+ on the wall patches. The layer stack targets $y^+ \approx 2.5$; the all- y^+ wall functions stay valid well beyond that, but values far above 30 mean the near-wall solution should not be trusted.

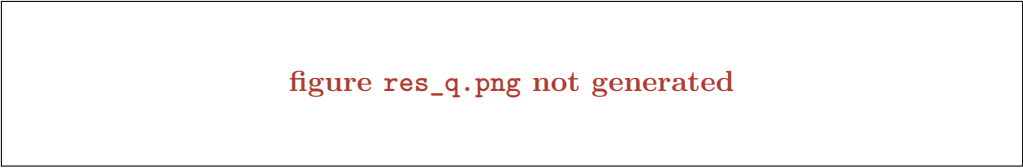


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Figure 21: Q-criterion isosurface, coloured by streamwise vorticity — the endplate and wheel vortex systems.